# Optimising Business Performance through Data Management and Analytics

# A Case Study on Sampat Kumar & Co

A Final report for the BDM Capstone Project

Submitted by

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# 1.Executive Summary

Sampat Kumar & Co. is a family-run wholesale and retail shop located in Visakhapatnam that sells food and packaging products. It serves both small businesses and regular customers. Even though the business has a good reputation, it faces several daily operational problems. These include excess stock that often remains unsold, risk of product expiry, limited storage space, and delays in billing during busy hours. These issues reduce efficiency and affect customer satisfaction as well as profits.

For this project, data was collected from three main sources: sales records, purchase records, and an inventory sheet that included opening stock, minimum stock levels, purchase rates, and expiry days. The data was cleaned and corrected to resolve unit differences and match stock levels with sales and purchases. Basic statistical analysis, stock balance checks, cost of goods sold (COGS)–based gross margin calculation, ABC analysis, and expiry risk analysis were carried out using Excel and Python.

The analysis showed that the business operates with a very low gross margin of about **3.99%**, which means profit per sale is very small. Many products were found to have large unsold stock, especially items with short expiry periods. This creates a high risk of hidden losses due to expiry and blocks a large amount of working capital. The ABC analysis also showed that only a few products contribute most of the sales value.

Based on these results, the study recommends better inventory control using ABC classification, FIFO for short-expiry items, and reducing purchase quantities for slow moving stock. These steps can help reduce wastage, improve storage usage, and increase overall business efficiency.

# 2. Detailed Explanation of Analysis Process/Method

### 2.1 Data Collection and Data Overview

The data used for this project was taken from the BUSY accounting and billing software used by Sampat Kumar & Co. The records were exported from BUSY into Excel format and then used for analysis. Three files were provided: Sales, Purchase, and Inventory.

The sales and purchase data cover the period from 1-11-2024 to 31-10-2025 and include transactions for 20 different items. The sales file contains details such as date, item name, quantity sold, and total sales value. The purchase file includes date, item name, quantity purchased, purchase rate, and total cost.

The inventory file is a one-time stock snapshot taken from the same system and includes item name, purchase rate, opening stock, minimum stock level, and expiry days. These three datasets together show the full flow of goods in the shop — from purchase, to storage, to final sale — and form the base for all further analysis.

### 2.2 Data Cleaning and Preprocessing

Before starting the final analysis, all three datasets (sales, purchase, and inventory) were carefully checked and cleaned. During the midterm review, it was clearly pointed out that the profit value I calculated earlier  AROUND 80 % was not realistic for a real shop. After receiving this feedback, I went back to the data, rechecked all calculations, and analysed the dataset again in detail to find where the mistake had happened.

First, all date columns in the sales and purchase data were converted into proper date format so that the data from 1-11-2024 to 31-10-2025 could be analysed correctly. Then all

quantity, rate, and amount columns were converted into numeric format so that total values and averages could be calculated without errors.

Next, the datasets were checked for missing values and duplicate entries. I did not find any major blank values. However, when I compared total sales with total purchases for each item, I noticed some logical issues. For example, for items like Almond and Mushroom, the total quantity sold was much higher than the quantity recorded as purchased. This showed that something was wrong with the stock data.

After this, I cleaned the item names across all three files. Some items had small spelling differences, and these were corrected so that the same product was not counted as different items.

A major issue was found in the inventory stock values. The opening stock and minimum stock levels were not written in actual units. After closely checking the numbers and discussing this with the shop staff, I understood that these stock values were recorded in hundreds. For example, an opening stock written as 113 actually meant 11,300 units. After confirming this, I multiplied all OPENING STOCK values by 100 to convert them into real units. After this correction, stock values such as Almond = 11,300 units, Sugar = 44,800 units, and Milk Powder = 24,300 units became realistic when compared with sales.

To finally check whether the data now made sense, I calculated a stock balance for each item using the formula:

$$\textit{Stock Balance} = \textit{Opening Stock} + \textit{Total Purchases} - \textit{Total Sales}$$

After this check, most items showed a positive balance, meaning that enough stock was available for sales. However, Almond (–2,877 units) and Mushroom (–1,029 units) still showed small negative balances. These were treated as missing opening stock at the start of the period and were mentioned clearly as a limitation instead of changing the original data

### 2.3 Item-wise Sales Performance Analysis

To understand how each product performed, the total sales value was calculated for every item. The sales data was grouped by item name, and the total sales amount was added for the full one-year period. This was done using Python and also checked using Excel pivot tables to make sure the numbers were correct.

This method helps to see which items bring more revenue and which items contribute less.
Business Problem Addressed:
The shop does not clearly know which products are most important for total revenue. This analysis helps identify high-selling and low-selling items so that stock and space can be managed better.

### 2.4 Cost of Goods Sold (COGS) and Gross Margin Calculation

To understand the real profit of the shop, the Cost of Goods Sold (COGS) was calculated. COGS was estimated by multiplying the total quantity sold by the average purchase rate of each item.
After that, gross profit was calculated by subtracting total cost from total sales value. Gross margin percentage was also calculated to understand how much profit the shop earns from its sales.
These calculations were done using the cleaned sales and purchase data in Python and Excel.
Business Problem Addressed:
Earlier, profit was estimated without using a proper cost method. This analysis helps measure the actual profit and understand whether the shop is earning enough margin.

### 2.5 Stock Balance Analysis

To understand how much stock remains after sales, a stock balance was calculated for each item using the formula:

**Stock Balance = Opening Stock + Total Purchases − Total Sales**

This calculation used corrected opening stock from the inventory file along with total purchase and total sales quantities from the respective datasets.

The purpose of this method was to check whether stock levels were balanced with actual demand.

Business Problem Addressed:
The shop faces issues of excess stock for some items and shortage for others. This analysis helps identify overstocked and understocked products so that future purchases can be planned properly.

### 2.6 Expiry Risk Analysis

Some products have limited shelf life. To understand expiry risk, the remaining stock was combined with the expiry days available for each item.
Items with high leftover stock and shorter expiry period were considered higher risk. Based on this, items were grouped into High, Medium, and Low risk categories.
This was done using Python and Excel.
Business Problem Addressed:
Some products get expired and cause loss to the shop. This analysis helps identify items that need quicker sales or smaller future orders.

### 2.7 ABC Analysis

ABC analysis was done to divide products based on their contribution to total sales value. Items were arranged from highest to lowest sales value, and cumulative percentage was calculated.

Based on their contribution, items were grouped into three categories:

- A category - most important items contributing highest share
- B category - moderately important items
- C category - less important items

This classification was done using Python and shown using charts.

Business Problem Addressed:

The shop currently gives equal attention to all items. ABC analysis helps focus more on important products and improve inventory management.

# 3. Results and Findings

### 3.1 Item wise Sales Performance

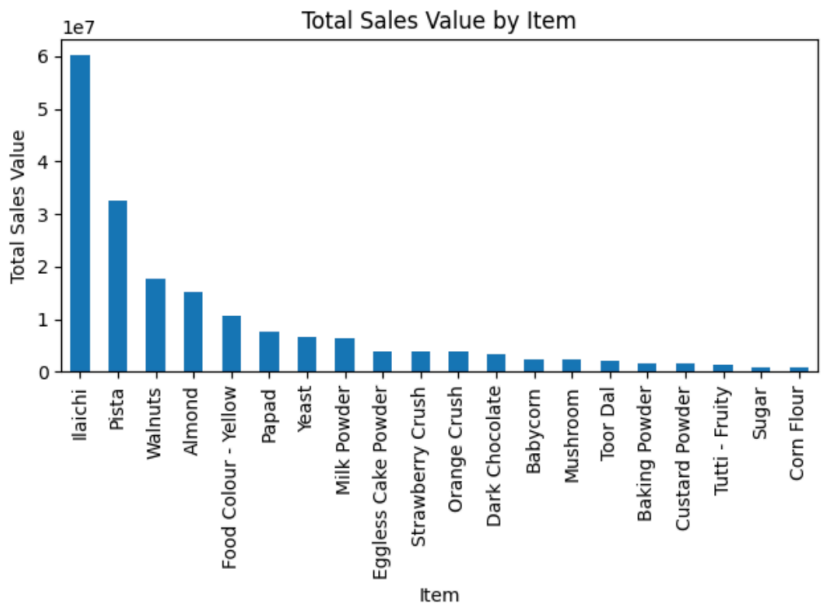


*Figure 3.1: Total Sales Value by Item*

From the chart, it is clear that only a few products generate a large portion of the total sales. Ilaichi is the highest revenue-generating item in the shop. Pista comes next and also

contributes a strong share, but it is lower than Ilaichi. These two items together form the top sales group and are very important for the shop's income.

Items like Walnuts, Almond, and Food Colour – Yellow fall in the middle category. They bring steady sales but their contribution is clearly lower than the top two items. Still, they remain important because together they add a reasonable share to the total revenue.

The remaining products such as Papad, Yeast, Eggless Cake Powder, Custard Powder, Tutti-Fruity, Sugar, Corn Flour, and Mushroom contribute smaller sales amounts. This shows a clear long-tail pattern, where many items sell in smaller quantities while only a few items generate most of the revenue.

Overall, the shop depends heavily on a small number of high-selling products. Ensuring regular availability of these items is very important for maintaining stable income.

### 3.2 Cost of Goods Sold (COGS) and Gross Margin

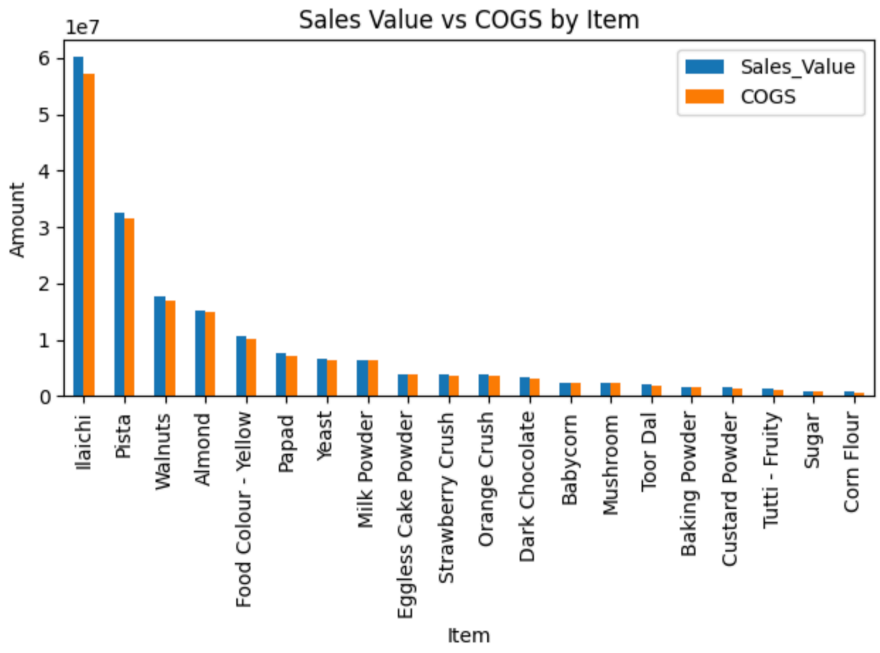


*Figure 3.2: Sales Value vs COGS by Item*

From Figure 3.2, it can be seen that items like Ilaichi and Pista have both high sales and high cost. These products generate large revenue, but they also require high purchase cost. Because of this, the difference between sales and cost is not very large.

For most products, the gap between sales value and COGS is relatively small. After calculating the overall gross margin, it was found to be around 3.99%. This means that even though the shop is selling good quantities, the actual profit per sale is quite low.

This indicates that improving cost control and reducing wastage are very important for increasing overall profit.

### 3.3 Stock Balance and Leftover Inventory

| Item | Opening_Stock_Corrected | Total_Purchased | Total_Sold | Stock_Balance |
|---|---|---|---|---|
| Almond | 11300 | 4952 | 19129 | -2877 |
| Babycorn | 30900 | 2787 | 18794 | 14893 |
| Baking Powder | 48200 | 2666 | 19032 | 31834 |
| Corn Flour | 44600 | 3930 | 18668 | 29862 |
| Custard Powder | 31900 | 4916 | 19164 | 17652 |
| Dark Chocolate | 23800 | 2747 | 18933 | 7614 |
| Eggless Cake Powder | 37700 | 3615 | 19031 | 22284 |
| Food Colour - Yellow | 49400 | 4420 | 19258 | 34562 |
| Ilaichi | 24400 | 4265 | 19097 | 9568 |
| Milk Powder | 24300 | 3958 | 19278 | 8980 |
| Mushroom | 15500 | 2228 | 18757 | -1029 |
| Orange Crush | 32000 | 2737 | 18893 | 15844 |
| Papad | 20000 | 3698 | 19025 | 4673 |
| Pista | 30200 | 3339 | 19419 | 14120 |
| Strawberry Crush | 23400 | 4449 | 19130 | 8719 |
| Sugar | 44800 | 3431 | 19244 | 28987 |
| Toor Dal | 33300 | 4700 | 18798 | 19202 |
| Tutti - Fruity | 34300 | 2657 | 19651 | 17306 |
| Walnuts | 24400 | 4367 | 18766 | 10001 |
| Yeast | 45800 | 3804 | 19049 | 30555 |

*Table 3.3: Sample Stock Balance for Selected Items*

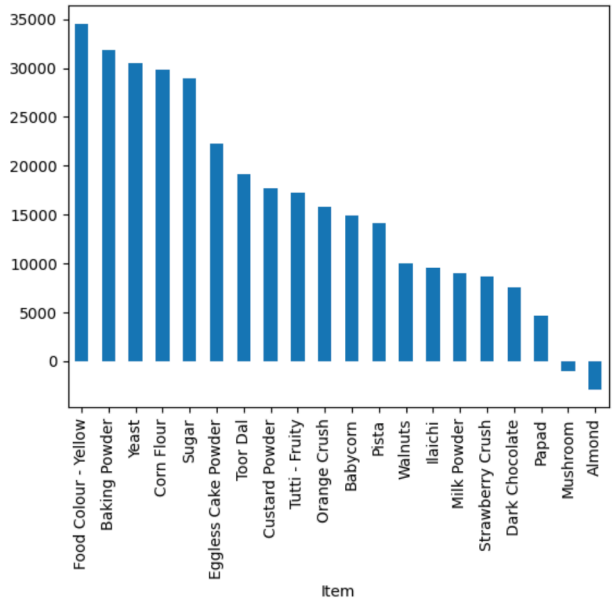


*Figure 3.3: Stock Balance by Item*

Table 3.3 shows the stock balance for selected items. Figure 3.3 shows the same data in a bar chart form. The chart makes it easier to understand how much stock is still left after sales.

From Table 3.3 and Figure 3.3, we can see that many items have very high leftover stock. Items like Food Colour – Yellow, Baking Powder, Yeast, Corn Flour, and Sugar still have very large quantities remaining. This shows that these products were purchased in higher quantities than what was actually needed based on sales.

Some other items such as Toor Dal, Tutti-Fruity, Custard Powder, and Orange Crush also have high unsold stock. Even though these products are selling, the quantity kept in stock is more than required. This can create storage problems and also increase the risk of expiry.

And also Almond and Mushroom show negative stock balance in Table 3.3 which means the recorded sales are more than the available stock. This might have happened because the opening stock was not recorded properly or because demand increased suddenly during the year.

Overall, both Table 3.3 and Figure 3.3 clearly show that purchase planning is not fully matched with actual demand. Because of this, some items are overstocked while a few are understocked. This can block working capital and reduce overall efficiency of the shop.

### 3.4 Expiry risk analysis

| Item | Opening_Stock_Corrected | Total_Purchased | Total_Sold | Stock_Balance | Expiry_Days |
|---|---|---|---|---|---|
| Almond | 11300 | 4952 | 19129 | -2877 | 57 |
| Babycorn | 30900 | 2787 | 18794 | 14893 | 99 |
| Baking Powder | 48200 | 2666 | 19032 | 31834 | 94 |
| Corn Flour | 44600 | 3930 | 18668 | 29862 | 157 |
| Custard Powder | 31900 | 4916 | 19164 | 17652 | 65 |
| Dark Chocolate | 23800 | 2747 | 18933 | 7614 | 164 |
| Eggless Cake Powder | 37700 | 3615 | 19031 | 22284 | 114 |
| Food Colour - Yellow | 49400 | 4420 | 19258 | 34562 | 76 |
| Ilaichi | 24400 | 4265 | 19097 | 9568 | 69 |
| Milk Powder | 24300 | 3958 | 19278 | 8980 | 134 |

*Table 3.5]4: Expiry Risk Level for Selected Items*

| Item | Stock_Balance | Expiry_Days | Expiry_Risk_Level |
|---|---|---|---|
| Almond | -2877 | 57 | Low Risk |
| Babycorn | 14893 | 99 | Medium Risk |
| Baking Powder | 31834 | 94 | Medium Risk |
| Corn Flour | 29862 | 157 | Low Risk |
| Custard Powder | 17652 | 65 | High Risk |
| Dark Chocolate | 7614 | 164 | Low Risk |
| Eggless Cake Powder | 22284 | 114 | Medium Risk |
| Food Colour - Yellow | 34562 | 76 | High Risk |
| Ilaichi | 9568 | 69 | Medium Risk |
| Milk Powder | 8980 | 134 | Medium Risk |

*Table 3.5: Stock Balance and Expiry Days*

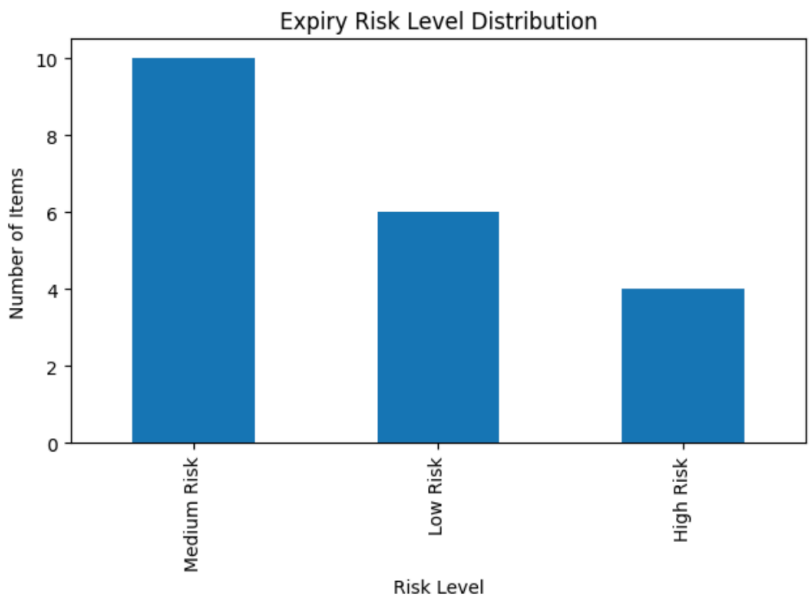


*Figure 3.4: Expiry Risk Level Distribution*

Some products in the shop have a limited shelf life. If they are not sold before their expiry date, they become waste and cause direct financial loss to the business.

Table 3.4 shows the remaining stock along with the expiry days for selected items. This table helps to see how much stock is left for each product and how many days are remaining before it expires.

When an item has high leftover stock and fewer expiry days, it becomes more risky. Based on this idea, the items were grouped into High, Medium, and Low expiry risk categories, as shown in Table 3.5. Figure 3.4 gives a simple visual view of how many items fall into each risk group.

From Table 3.5 and Figure 3.4, it can be observed that items like Yeast, Custard Powder, Baking Powder, and Mushroom fall under the high-risk category. These products have large remaining stock and shorter expiry periods. If they are not sold quickly, they may expire and result in loss.

Items such as Milk Powder, Papad, and Tutti-Fruity fall under the medium-risk category. These items are not in immediate danger, but they still need regular monitoring and careful ordering.

The remaining items fall into the low-risk group because their stock level is more balanced or their expiry period is longer.

Overall, this analysis shows that the shop needs better control over short-expiry items. If proper attention is given, expiry loss can be reduced and profit can be protected.

### 3.5 ABC Analysis

| Item | Sales_Value | Cumulative_Sales | Cumulative_% | ABC_Category |
|---|---|---|---|---|
| Ilaichi | 60155550 | 60155550 | 32.501869 | A |
| Pista | 32623920 | 92779470 | 50.128478 | A |
| Walnuts | 17827700 | 110607170 | 59.760732 | A |
| Almond | 15303200 | 125910370 | 68.029007 | A |
| Food Colour - Yellow | 10591900 | 136502270 | 73.751779 | B |
| Papad | 7610000 | 144112270 | 77.863440 | B |
| Yeast | 6667150 | 150779420 | 81.465682 | B |
| Milk Powder | 6361740 | 157141160 | 84.902912 | B |
| Eggless Cake Powder | 3996510 | 161137670 | 87.062214 | B |
| Strawberry Crush | 3826000 | 164963670 | 89.129391 | B |
| Orange Crush | 3778600 | 168742270 | 91.170958 | C |
| Dark Chocolate | 3313275 | 172055545 | 92.961111 | C |
| Babycorn | 2443220 | 174498765 | 94.281175 | C |
| Mushroom | 2438410 | 176937175 | 95.598641 | C |
| Toor Dal | 2067780 | 179004955 | 96.715857 | C |
| Baking Powder | 1598688 | 180603643 | 97.579623 | C |
| Custard Powder | 1533120 | 182136763 | 98.407963 | C |
| Tutti - Fruity | 1277315 | 183414078 | 99.098093 | C |
| Sugar | 885224 | 184299302 | 99.576377 | C |
| Corn Flour | 784056 | 185083358 | 100.000000 | C |

*Table 3.6: ABC Classification Based on Sales Value*

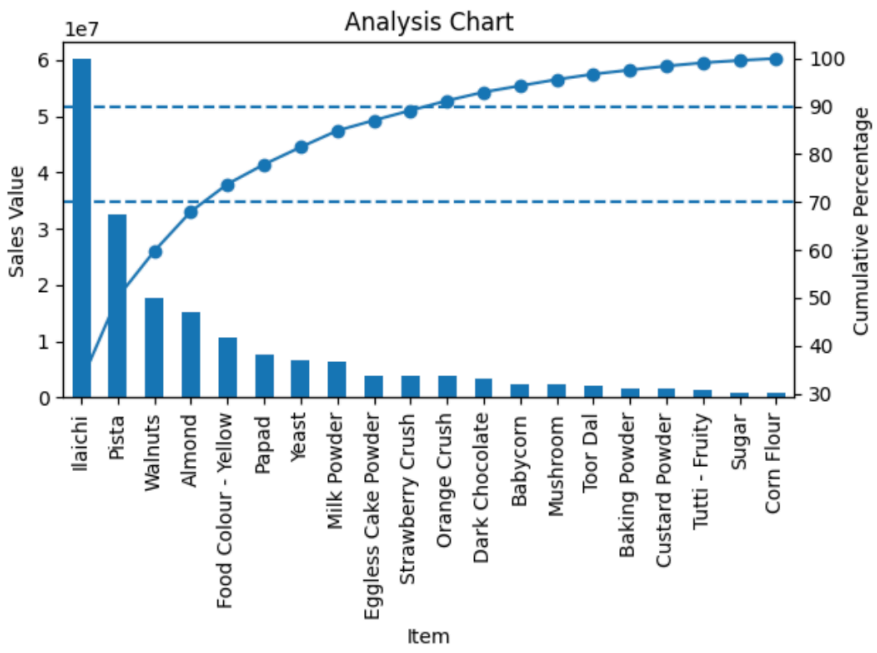


*Figure 3.5: ABC Analysis Chart (Sales Value and Cumulative Percentage)*

From the ABC table, only a few products fall under the A category, but they contribute the highest share of total revenue. Items such as Ilaichi and Pista belong to this group and are extremely important for the business. Any issue with these products, such as stock-out or pricing error, can affect overall revenue significantly.

The B category contains items that provide moderate contribution. These products are important but not as critical as A items.

Most remaining items fall under the C category. These products contribute only a small share of total sales. Even though they are many in number, their overall impact on revenue is limited.

This clearly shows that the shop's income mainly depends on a small group of key products. Proper focus on A items, controlled management of B items, and careful ordering of C items can improve overall inventory efficiency.

# 4. Interpretation of Results and Recommendations:

## 4.1 Interpretation of Results

### 4.1.1 Sales and Profit Interpretation

From the sales and COGS analysis calculations, the shop is operating at a gross margin of only about 4%. In practical terms, this means the shop sells a good volume of products, but earns only a small profit on each sales. A large part of the revenue comes from a few high value products such as Ilaichi and Pista. Because of this, even a small change in the purchase price, wastage, or selling price of these key items can have a impact on the overall profit.

### 4.1.2 Stock Balance Interpretation

From the stock balance analysis, I found that many items have a lot of stock still left in the shop. For example, Food Colour, Baking Powder, Yeast, Sugar, and Corn Flour all have very high closing stock. This suggests that these items were bought in larger quantities than needed.
On the other hand, some items like Almond and Mushroom show a negative stock balance. This means that the recorded sales are more than the total of opening stock plus purchases. This could be because some opening stock was not recorded properly or there were mistakes in data entry. These items need to be checked again physically and corrected in the records.
Overall, this mismatch between leftover stock and actual demand shows that purchase planning is not fully based on sales data. As a result, money gets stuck in extra stock, storage space is used inefficiently, and there is a higher chance of wastage.

### 4.1.3 Expiry Risk Interpretation

The expiry risk analysis shows that some items are in the very high risk group. These items have a lot of unsold stock and also a short remaining shelf life. Examples are Yeast, Custard Powder, Baking Powder, and Mushroom. If these products are not sold quickly, they may get expired and then cannot be sold. This will lead to a loss for the shop.

### 4.1.4 ABC Analysis Interpretation

From the ABC analysis, I observed that only a few items in the A category contribute most of the total sales value. Items like Ilaichi and Pista come under this group and are very important for the business.

Many other products fall into the C category. These items bring in only a small part of the revenue, but they still take up space and need time for handling and billing.

This suggests that the shop is spending effort on many low-value items while its main income actually comes from a small group of key products.

## 4.2 Recommendations

Using the above results, I have listed some recommendations for the shop. I have divided them into short-term actions that can be done soon and long-term actions that need continuous effort or implementation.

### 4.2.1 Urgent (Short Term) Recommendations

1. Reduce Over Stocked Items

Items that have very high leftover stock, such as Baking Powder, Yeast, Food Colour, Sugar, and Corn Flour, should be ordered in smaller quantities for the next few purchase cycles.

This will reduce extra stock, and release money that is currently tied up in the inventory, and reduce the chance of products expiring on the shelf.

2. Clear High Expiry Risk Stock Quickly

Items with high expiry risk, such as Yeast, Custard Powder, and Mushroom, should be given priority for sale. The shop can use simple methods like small discounts, combo packs, or putting these items in a more visible place. The aim is to sell them before they expire so that wastage is reduced

3. Focus Extra on A-Category Items

The A category items like Ilaichi and Pista should almost never go out of stock, because they include large part of the total sales. The shop should regularly monitor their stock levels and prices. Even a small mistake in these products can have a big effect on overall revenue.

4. Correct Stock Recording for Negative Balance Items

For items such as Almond and Mushroom, which showed a negative stock balance, the shop should do a physical stock check. After counting the actual quantity, the records in the system should be updated. This will help avoid similar data problems in future analysis.

### 4.2.2 Long-Term Recommendations

1. Adopt Demand-Based Purchasing

In the long run, the shop should plan its purchases based on past sales data. Instead of buying large quantities without checking the demand, it can look at how much of each item was sold in the last few months and then decide the next purchase quantity. This will reduce both over and under stocking.

2. Regular Stock Review System

The shop should fix a regular schedule for stock review, for example once every week or once every month. During this check, they can note which items are selling slowly and which are moving fast. This will help them react early, instead of waiting until the problem becomes big.

3. Follow ABC-Based Inventory Control

The shop can use the ABC categories to decide how much attention to give each product group:

- For A items they can check stock and price frequently and try to avoid out of stock as they most crucial for the part of revenue.
- For B items they have to monitor regularly but with normal control.
- For C items they need to order in smaller quantities and give them less shelf space so that they do not crash.

4. Expiry Tracking System

The shop should maintain a simple expiry tracking system. For example, they can keep a list or mark items with stickers showing the expiry month. Products that are close to expiry should be moved to the front and promoted so that they get sold first.

5. Improve Profit Margin Through Better Pricing and Cost Control

Because the gross margin is quite low around 4 percent, the shop should try to slightly improve its profit in the following ways:

- Talk to suppliers and try to get a better purchase rate which is low purchase rate, especially for high sales items.
- Reduce wastage by controlling extra stock and managing expiry better.
- Check or review the selling prices of popular items and check if a small increase is possible without losing customers.

## 4.3 Expected Impact of Implementing These Recommendations

If the shop follows these suggestions, it can expect several benefits:

- Fewer products will expire on the shelf, so loss due to expiry will reduce.
- Less money will be used in extra stock, which will improve cash flow.
- Profit or gross may improve slowly as wastage and stock mistakes reduce.
- Inventory will be better organised and easier to track.
- Important fast moving items will be more regularly available for customers.
- Customers are likely to have a smoother experience because items are in stock and billing delays may reduce.

LINKS:

**DATASET LINK:** https://docs.google.com/spreadsheets/d/1PjVvWw0WZeHUhymucg8ZiikGQ4QBzLKf/edit?usp=sharing&ouid=117600913125759666117&rtpof=true&sd=true

**COLAB LINK:** **HTTPS://COLAB.RESEARCH.GOOGLE.COM/DRIVE/1WVC3TDQSKEXFPF8CJ9PBCFQTWUPFDJR8?USP=SHARING**

DRIVE LINK: HTTPS://DRIVE.GOOGLE.COM/DRIVE/U/1/FOLDERS/1KMMLXAA7RFATOM4VYTSO95YD9SEW6MA5